

8.1 Apply the Pythagorean Theorem

MPM1D

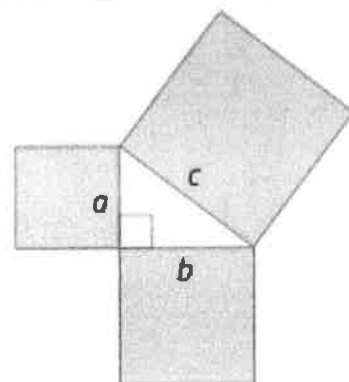
Before we can apply the Pythagorean theorem, let's review what we know about right triangles.

DO IT NOW!!!!

A right triangle is a triangle that has one angle equal to 90 degrees.

The hypotenuse of a right triangle is:

- i) The longest side of a right triangle
- ii) The side opposite the right angle



The Pythagorean theorem states: In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two shorter sides.

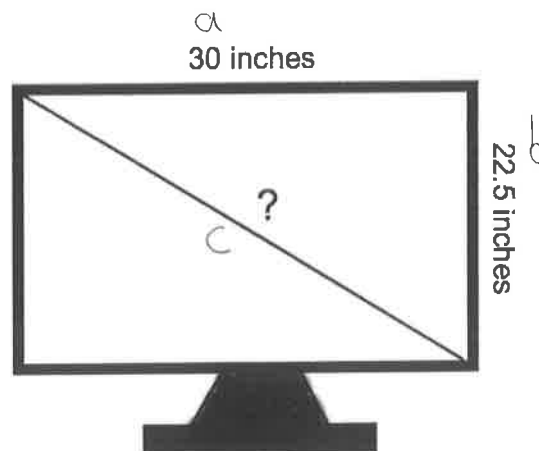
$$a^2 + b^2 = c^2 \quad \text{or} \quad c^2 = a^2 + b^2$$

In a right triangle, we use the letters a and b to represent the two shorter sides (the legs) and the letter c to represent the longest side (the hypotenuse).

Part A: Use Pythagorean Theorem to Find the Hypotenuse

Example 1: The advertised size of a computer or television screen is actually the length of the diagonal of the screen. A television screen measures 30 inches by 22.5 inches. Determine the length of its diagonal.

$$\begin{aligned} a^2 + b^2 &= c^2 \\ (30)^2 + (22.5)^2 &= c^2 \\ 1406.25 &= c^2 \\ \sqrt{1406.25} &= c \\ c &= 37.5 \end{aligned}$$



37.5 inches

Example 2: When Laurie swims laps in her rectangular pool, she decides to swim along the diagonal, so that she does not have to turn around as often. Find the distance Laurie travels by swimming once along the diagonal. Round your answer to the nearest tenth.

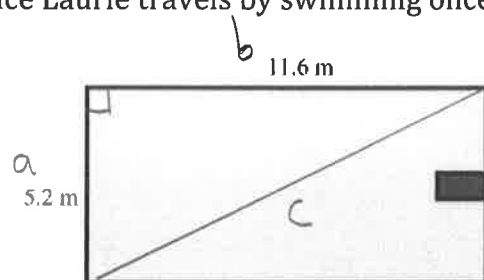
$$a^2 + b^2 = c^2$$

$$(5.2)^2 + (11.6)^2 = c^2$$

$$161.6 = c^2$$

$$\sqrt{161.6} = c$$

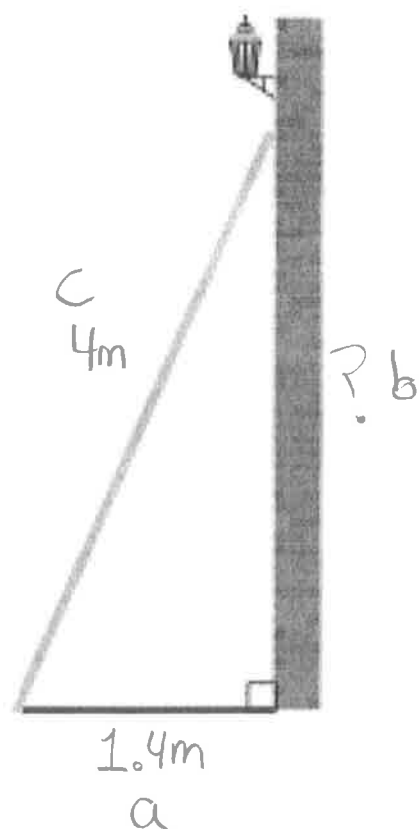
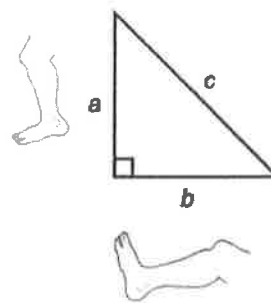
$$c = 12.7 \text{ m}$$



12.7 m

Part B: Use Pythagorean Theorem to Find a Leg

Example 3: Jenna is changing a light bulb. She rests a 4 meter ladder against a vertical wall so that its base is 1.4 meters from the wall. How high up the wall does the top of the ladder reach? Round to the nearest tenth of a meter.



$$a^2 + b^2 = c^2$$

$$(1.4)^2 + b^2 = (4)^2$$

$$b^2 = (4)^2 - (1.4)^2$$

$$b = \sqrt{14.04}$$

$$b = 3.7$$

It reaches 3.7 m
up the wall

Part C: Apply the Pythagorean Theorem

Remember:

$$\text{Area of a triangle} = \frac{1}{2}(\text{base})(\text{height})$$

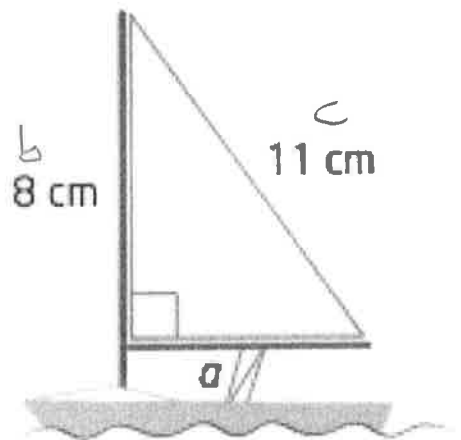
and

the legs of a right triangle are the base and height

Example 4: Calculate the area of the triangular sail on the toy sailboat.

Step 1: Find the length of side a (this is the base of the triangle)

$$\begin{aligned}a^2 + b^2 &= c^2 \\a^2 &= c^2 - b^2 \\a^2 &= (11)^2 - (8)^2 \\a^2 &= 57 \\a &= \sqrt{57} \\a &= 7.5 \text{ cm}\end{aligned}$$



Step 2: Find the area of the sail

$$\begin{aligned}\text{base} &= 7.5 \\ \text{height} &= 8 \\ A &= \frac{1}{2}bh \\ A &= \frac{1}{2}(7.5)(8) \\ &= 30 \text{ cm}^2\end{aligned}$$

The area is 30 cm^2

Example 5: Zeke drives from his house in Geometropolis to his cottage in Decimal Point. He drives 64 km east and then 135 km north. How much travel distance will he save when he drives along the new Square Route Expressway?

Distance along Expressway:

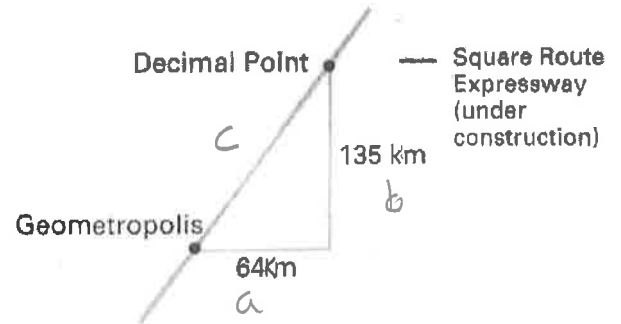
$$a^2 + b^2 = c^2$$

$$(64)^2 + (135)^2 = c^2$$

$$22\,321 = c^2$$

$$\sqrt{22\,321} = c$$

$$c = 149.4 \text{ km}$$



$$\begin{aligned} \text{Distance saved} &= (64 + 135) - 149.4 \\ &= 49.6 \text{ km} \end{aligned}$$

He will save 49.6 km

Homework: Complete Worksheet